

# USB4 2.0 ENGINEERING CHANGE NOTICE FORM

**Title: IRL Limits Adjustment**

**Applied to: USB4 Specification Version 2.0**

|                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Brief description of the functional changes:</b>                                                                                                                                                                           |
| Change the Tx IRL limit from $-15 + (\text{Tx\_ISI\_Margin} - 11.5) \times 0.57$ to $-13.5 + (\text{Tx\_ISI\_Margin} - 11.5) \times 0.375$ (normative)<br>Change the Rx IRL from -14.5dB (normative) to -13.5dB (informative) |

|                                                                                                |
|------------------------------------------------------------------------------------------------|
| <b>Benefits as a result of the changes:</b>                                                    |
| Relax the IRL limits which were identified as too aggressive causing high “false failure” rate |

|                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------|
| <b>An assessment of the impact to the existing revision and systems that currently conform to the USB specification:</b> |
| None                                                                                                                     |

|                                                  |
|--------------------------------------------------|
| <b>An analysis of the hardware implications:</b> |
| None                                             |

|                                                  |
|--------------------------------------------------|
| <b>An analysis of the software implications:</b> |
| None                                             |

|                                                            |
|------------------------------------------------------------|
| <b>An analysis of the compliance testing implications:</b> |
| None                                                       |

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## Actual Change

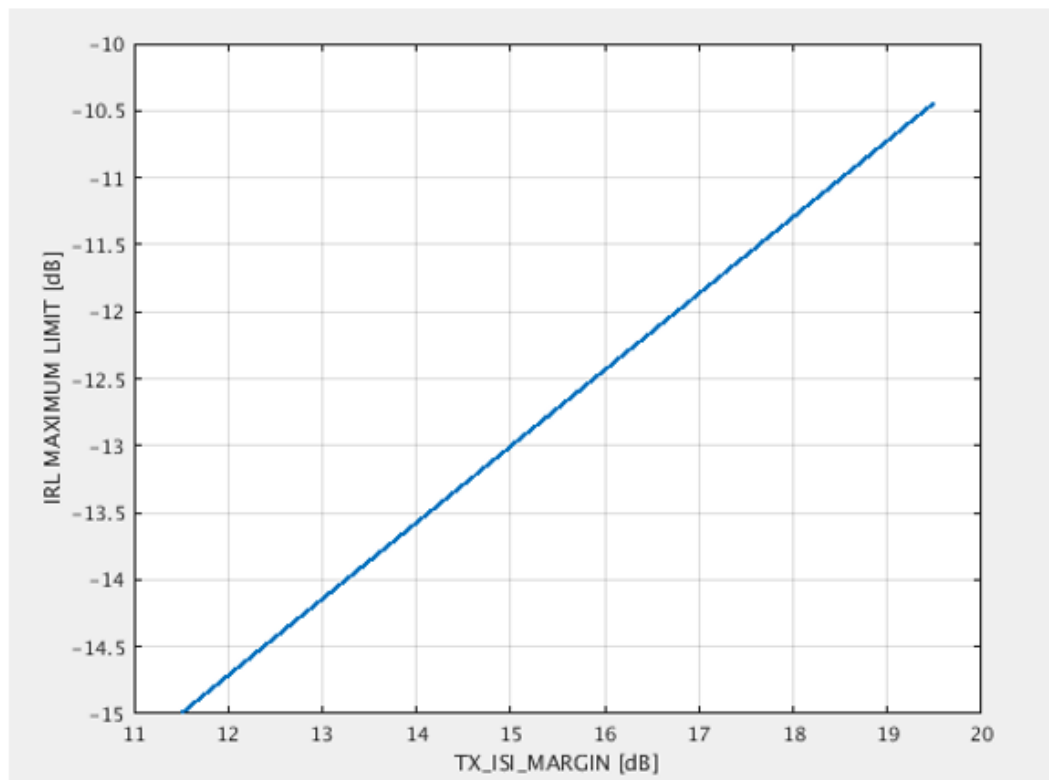
### (a). Section 3.2.3.6

#### From Text:

The TX\_IRL maximum limit is a function of the measured TX\_ISI\_MARGIN, as specified in Section 3.2.3.5:

$$IRL \leq -15 + (TX\_ISI\_MARGIN - 11.5) \times 0.57$$

**Figure 3-30. IRL Maximum Limit as a Function of TX\_ISI\_MARGIN**



#### To Text:

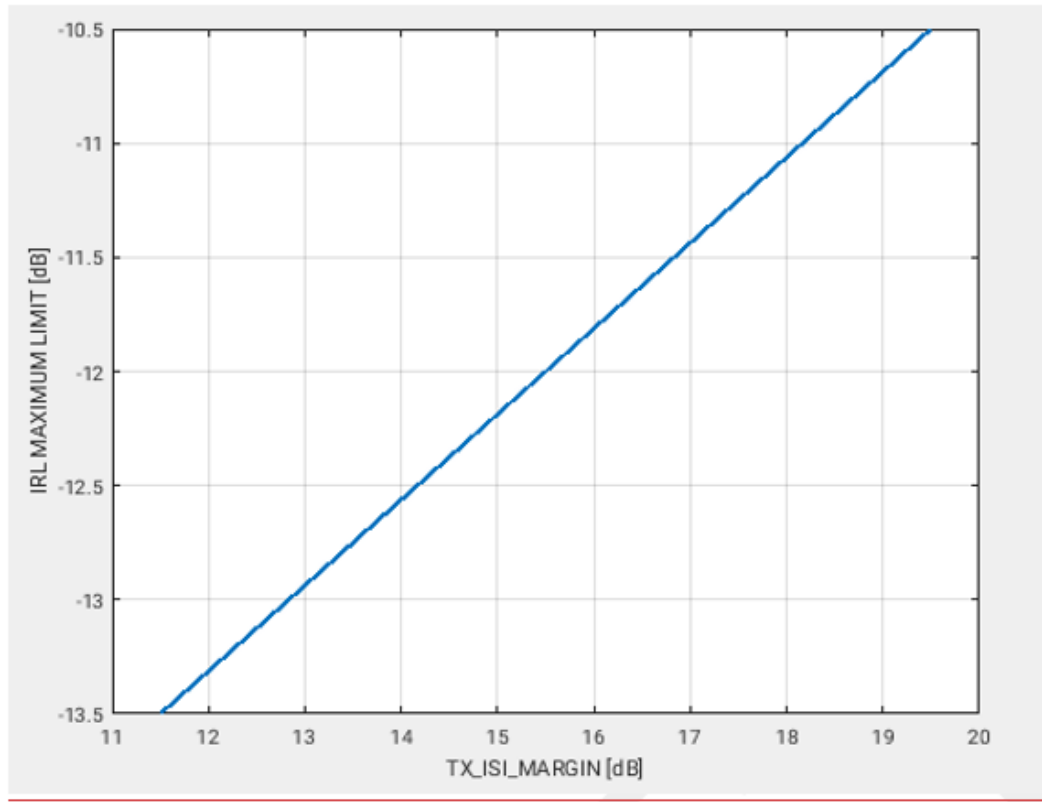
The TX\_IRL maximum limit is a function of the measured TX\_ISI\_MARGIN, as specified in Section 3.2.3.5:

$$IRL \leq -15 + (TX\_ISI\_MARGIN - 11.5) \times 0.57$$

$$IRL \leq -13.5 + (TX\_ISI\_MARGIN - 11.5) \times 0.375$$

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**Figure 3-30. IRL Maximum Limit as a Function of TX\_ISI\_MARGIN**



## (b). Section 3.2.4.1 (Table 3-26)

### From Text:

**Table 3-26. Receiver Specifications**

| Symbol            | Description                                              | Min | Max                   | Units | Comments              |
|-------------------|----------------------------------------------------------|-----|-----------------------|-------|-----------------------|
| RX_TOLERANCE      | Receiver tolerance testing                               |     |                       |       | See Section 3.2.4.2   |
| RX_IRL            | Integrated Return-Loss at TP3'                           |     | -14.5                 | dB    | See Section 3.2.4.1.1 |
| RL (Informative)  | Informative Return Loss Mask at TP3'                     |     | See Section 3.2.4.1.2 | dB    |                       |
| LANE_TO_LANE_SKEW | Skew between incoming signals of the same Port at TP3'   |     | 44                    | ns    | See Note 1.           |
| V_MAX             | Input signal peak differential voltage tolerance at TP3' |     | 600                   | mV    | See Note 4.           |

### To Text:

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**Table 3-26. Receiver Specifications**

| Symbol               | Description                                              | Min | Max                   | Units | Comments              |
|----------------------|----------------------------------------------------------|-----|-----------------------|-------|-----------------------|
| RX_TOLERANCE         | Receiver tolerance testing                               |     |                       |       | See Section 3.2.4.2   |
| RX_IRL_(Informative) | <u>Informative</u> Integrated Return-Loss at TP3'        |     | - <del>14</del> 13.5  | dB    | See Section 3.2.4.1.1 |
| RL (Informative)     | Informative Return Loss Mask at TP3'                     |     | See Section 3.2.4.1.2 | dB    |                       |
| LANE_TO_LANE_SKEW    | Skew between incoming signals of the same Port at TP3'   |     | 44                    | ns    | See Note 1.           |
| V_MAX                | Input signal peak differential voltage tolerance at TP3' |     | 600                   | mV    | See Note 4.           |